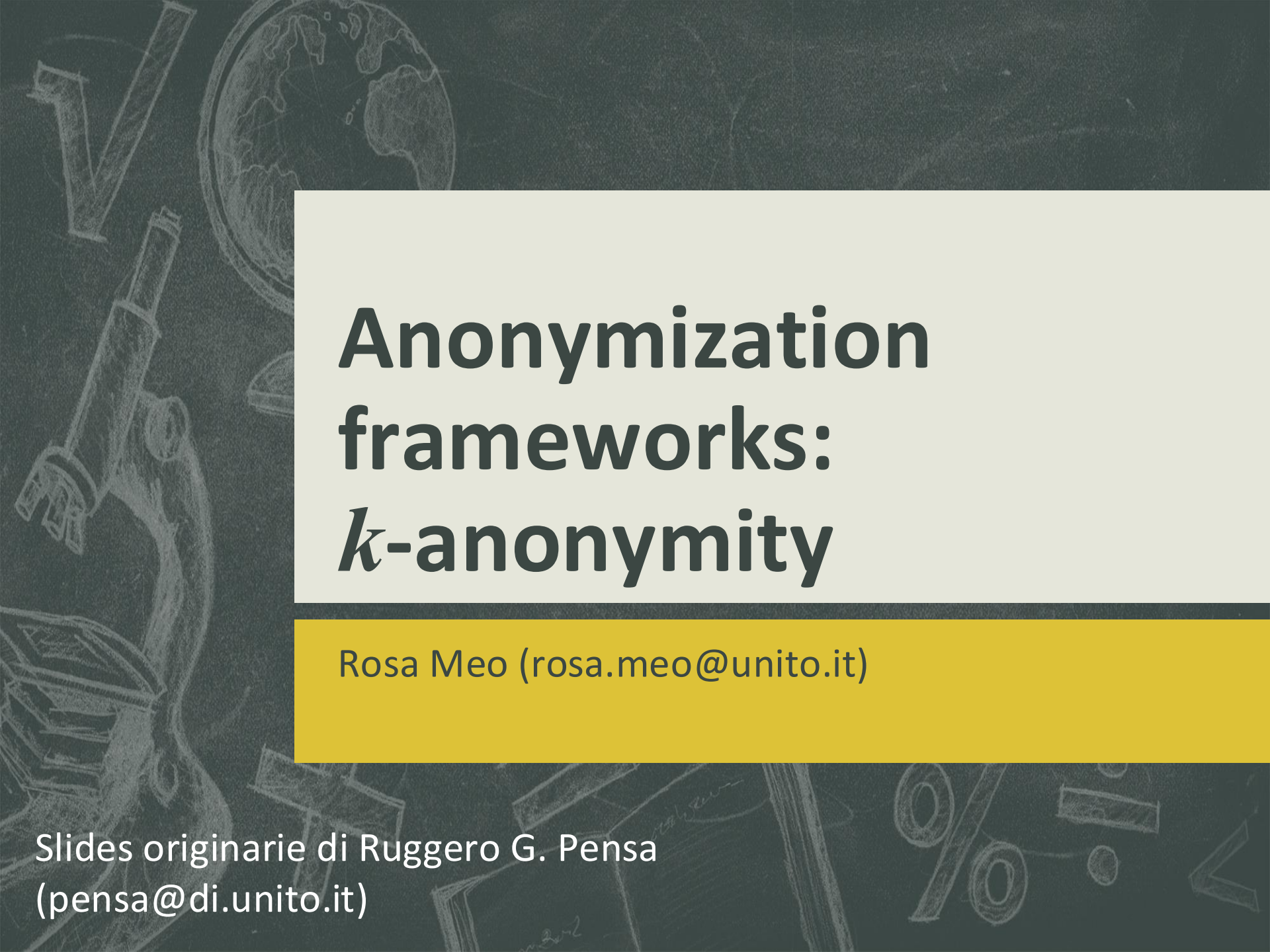




Anonymization frameworks: *k*-anonymity

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Slides originarie di Ruggero G. Pensa
(pensa@di.unito.it)

The anonymity problem

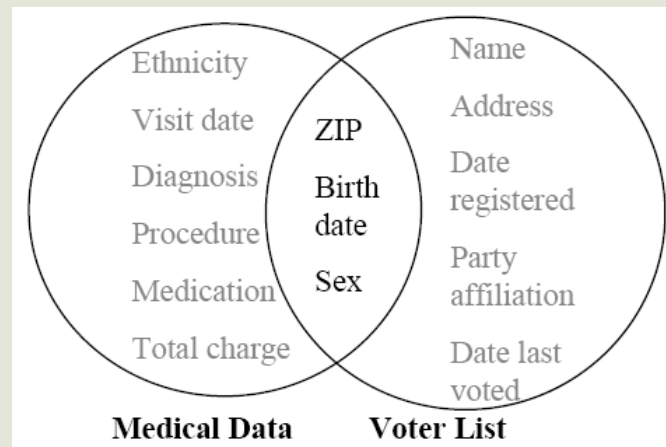
- Several agencies, institutions, bureaus, organizations make (sensitive) data involving people publicly available
 - termed **microdata** used for analysis
 - often required and imposed by law
- To protect privacy microdata are **sanitized**
 - explicit identifiers (SSN, name, phone #) are removed
- Not sufficient for preserving privacy
 - susceptible to **linking attacks**
 - publicly available databases (voter lists, city directories) can reveal the “hidden” identity

Linking attack



In 2001, Latanya Sweeney managed to re-identify the medical record of the governor of Massachusetts

- MA collects and publishes sanitized medical data for state employees (microdata) **left circle**
- voter registration list of MA (publicly available data) **right circle**



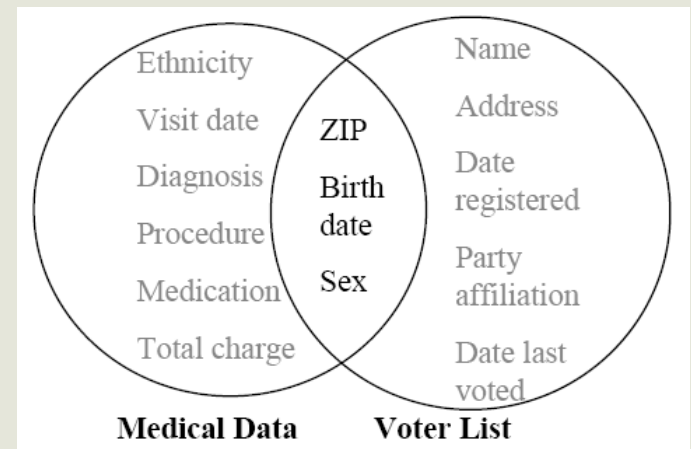
Linking attack

Looking for governor's record

- join the tables:
 - 6 people had his birth date
 - 3 were men
 - 1 in his zipcode

Regarding the US 1990 census data

- 87% of the population are unique based on
 - Zipcode
 - Gender
 - Date of birth



Linking attack

Massachusetts hospital discharge dataset

Medical Data Released as Anonymous

SSN	Name	Ethnicity	Date Of Birth	Sex	ZIP	Marital Status	Problem
		asian	09/27/64	female	02139	divorced	hypertension
		asian	09/30/64	female	02139	divorced	obesity
		asian	04/18/64	male	02139	married	chest pain
		asian	04/15/64	male	02139	married	obesity
		black	03/13/63	male	02138	married	hypertension
		black	03/18/63	male	02138	married	shortness of breath
		black	09/13/64	female	02141	married	shortness of breath
		black	09/07/64	female	02141	married	obesity
		white	05/14/61	male	02138	single	chest pain
		white	05/08/61	male	02138	single	obesity
		white	09/15/61	female	02142	widow	shortness of breath

Voter List

Name	Address	City	ZIP	DOB	Sex	Party	*****
.....							
.....							
Sue J. Carlson	1459 Main St.	Cambridge	02142	9/15/61	female	democrat	
.....							

Figure 1: Re-identifying anonymous data by linking to external data

Public voter dataset

Re-identification

In 2000, the US population was uniquely identifiable by

- year of birth, 5-digit ZIP code: **0.2%**
- Year of birth, country: **0.0%**
- Year and month of birth, 5-digit ZIP code: **4.2%**
- Year and month of birth, country: **0.2%**
- Year, month and day of birth, 5-digit ZIP code: **63.3%**
- Year, month and date of birth, country: **14,8%**

Privacy in microdata

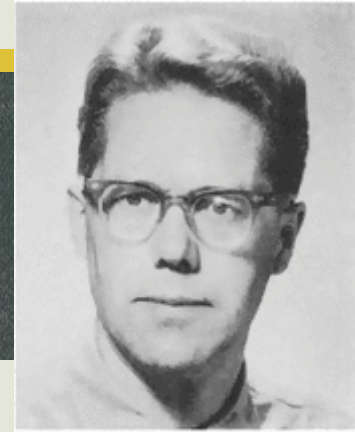
Role of attributes in microdata

- **explicit identifiers** are removed
- **quasi identifiers** can be used to re-identify individuals
- **sensitive attributes** (may not exist!) carry sensitive information

identifier	quasi identifiers			sensitive
Name	Birthdate	Sex	Zipcode	Disease
Andre	21/1/79	male	53715	Flu
Beth	10/1/81	female	55410	Hepatitis
Carol	1/10/44	female	90210	Brochitis
Dan	21/2/84	male	02174	Sprained Ankle
Ellen	19/4/72	female	02237	AIDS

goal of privacy preservation (rough definition)
de-associate individuals from **sensitive information**

Quasi-identifiers



- Quasi-identifiers are **attributes** that are not of themselves unique identifiers, but can be **combined with other quasi-identifiers** to create a **unique identifier** [Tore Dalenius, 1986]
- Quasi-identifiers have been the basis of several attacks on released data:
 - Sweeney linked health records to publicly available information to locate the then-governor of Massachusetts' hospital records
 - Sweeney, Abu and Winn used public voter records to re-identify participants in the Personal Genome Project
 - Arvind Narayanan and Vitaly Shmatikov made use of quasi-identifiers to de-anonymize data released by Netflix

k -anonymity



k -anonymity, together with its enforcement via **generalization** and **suppression**, has been proposed as an approach to protect respondents' identities while releasing truthful information

- It was first proposed Pierangela Samarati and Latanya Sweeney in 1998
- Then it was developed by the two scientists independently

Quasi-identifier (definition)

Let $A = \{a_1, \dots, a_n\}$ be a set of n attributes and D a dataset (table) defined over A

Definition 1: Quasi-identifier

A **quasi-identifier** of D is a set of attributes $QI \subseteq A$ whose release must be controlled.

Example

N	SSN	Name	Ethnicity	Date of Birth	Gender	ZIP	Marital Status	Problem
1			AfrAm	09/27/64	Male	02139	Divorced	Obesity
2			AfrAm	09/30/64	Male	02139	Divorced	Hypertension
3			AfrAm	04/18/64	Male	02139	Married	Chest pain
4			AfrAm	04/15/64	Male	02139	Married	Chest pain
5			AfrAm	09/15/64	Male	02138	Married	Shortness of breath
6			Caucasian	03/13/63	Male	02141	Married	Hypertension
7			Caucasian	03/18/63	Male	02141	Married	Shortness of breath
8			Caucasian	09/13/64	Female	02138	Married	Shortness of breath
9			Caucasian	06/07/64	Female	02138	Married	Obesity
10			Caucasian	05/14/61	Female	02138	Single	Chest pain
11			Caucasian	05/08/61	Female	02138	Single	Obesity

Example

N	SSN	Name	Ethnicity	Date of Birth	Gender	ZIP	Marital Status	Problem
1			AfrAm	09/27/64	Male	02139	Divorced	Obesity
2			AfrAm	09/30/64	Male	02139	Divorced	Hypertension
3			AfrAm	04/18/64	Male	02139	Married	Chest pain
4			AfrAm	04/15/64	Male	02139	Married	Chest pain
5			AfrAm	09/15/64	Male	02138	Married	Shortness of breath
6			Caucasian	03/13/63	Male	02141	Married	Hypertension
7			Caucasian	03/18/63	Male	02141	Married	Shortness of breath
8			Caucasian	09/13/64	Female	02138	Married	Shortness of breath
9			Caucasian	06/07/64	Female	02138	Married	Obesity
10			Caucasian	05/14/61	Female	02138	Single	Chest pain
11			Caucasian	05/08/61	Female	02138	Single	Obesity

2-anonymous w.r.t. {Ethnicity, Gender, Marital Status}

1-anonymous w.r.t. {Ethnicity, Gender, Marital Status, **ZIP**}

1-anonymous w.r.t. {Date of Birth}

k -anonymity (definition)

Let T be a dataset over the set $A = \{a_1, \dots, a_n\}$ of attributes and QI_T the set of quasi-identifiers associated to T .

T **satisfies k -anonymity** iff for each quasi-identifier $QI \in QI_T$ each existing sequence of values of attributes QI appears at least with k occurrences in T .

k -anonymity

Each release of data must be such that every combination of values of quasi-identifiers can be indistinctly matched to at least k respondents

- hide each individual among **$k-1$** others
- linking cannot be performed with confidence **$> 1/k$**

k -anonymity requires that, in the released table itself, the respondents be indistinguishable (within a given set) with respect to a set of attributes

Sufficient condition for the k -anonymity requirement:

- To guarantee the k -anonymity requirement, each quasi-identifier set of values in the released table must have at least k occurrences
- **(sensitive attributes are not considered)**

Example

N	SSN	Name	Ethnicity	Date of Birth	Gender	ZIP	Marital Status	Problem
1			AfrAm	09/27/64	Male	02139	Divorced	Obesity
2			AfrAm	09/30/64	Male	02139	Divorced	Hypertension
3			AfrAm	04/18/64	Male	02139	Married	Chest pain
4			AfrAm	04/15/64	Male	02139	Married	Chest pain
5			AfrAm	09/15/64	Male	02138	Married	Shortness of breath
6			Caucasian	03/13/63	Male	02141	Married	Hypertension
7			Caucasian	03/18/63	Male	02141	Married	Shortness of breath
8			Caucasian	09/13/64	Female	02138	Married	Shortness of breath
9			Caucasian	06/07/64	Female	02138	Married	Obesity
10			Caucasian	05/14/61	Female	02138	Single	Chest pain
11			Caucasian	05/08/61	Female	02138	Single	Obesity

2-anonymous w.r.t. {Ethnicity, Gender, Marital Status}

1-anonymous w.r.t. {Ethnicity, Gender, Marital Status, **ZIP**}

1-anonymous w.r.t. {Date of Birth}

How to achieve k -anonymity

Generalization: the values of a given attribute are replaced by using more general values

- Based on the definition of generalization hierarchy
- Examples:
 - ZIP codes replaced by hiding the two least significant digits (10149 → 101**, 10126 → 101**)
 - Date of birth replaced by year of birth (09/27/1964 → 1964, 09/30/1964 → 1964)

Suppression: it is a well-known technique consisting in protecting sensitive information by removing it

- The introduction of suppression can reduce the amount of generalization necessary to satisfy the k -anonymity constraint

k -anonymity example

Generalization

- publish more general values, i.e., given a domain hierarchy, roll-up

Suppression

- remove tuples, i.e., do not publish outliers

Birthdate	Sex	Zipcode
21/1/79	male	53715
10/1/79	female	55410
1/10/44	female	90210
21/2/83	male	02274
19/4/82	male	02237

original microdata

	Birthdate	Sex	Zipcode
group 1	*/1/79	person	5****
	*/1/79	person	5****
suppressed	1/10/44	female	90210
group 2	*/*/8*	male	022**
	//8*	male	022**

2-anonymous data

It is not easy to list sensitive data and the quasi identifiers

- There are "proxy" attributes that are related to sensitive data

Sensitive Variable	Example Proxies
Gender	Education Level, Income, Occupation, Felony Data, Keywords in User Generated Content (e.g. CV, Social Media etc.), University Faculty, Working Hours
Marital Status	Education Level, Income
Race	Felony Data, Keywords in User Generated Content (e.g. CV, Social Media etc.), Zipcode
Disabilities	Personality Test Data

- From: Simon Caton and Christian Haas (2020) FAIRNESS IN MACHINE LEARNING: A SURVEY, in arXiv: 2010:04053v1, 4 October 2020

Privacy vs. Utility

Anonymized data are released for statistical analysis purposes

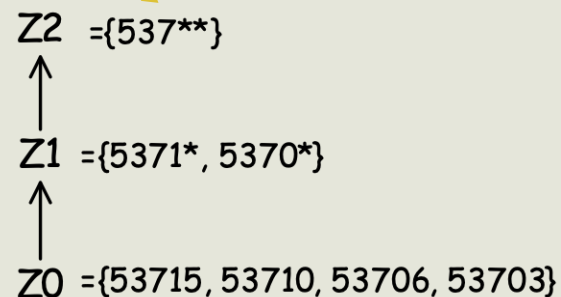
If data are “too anonymous” they become useless

Do not anonymize more than necessary!

Domain generalization hierarchies

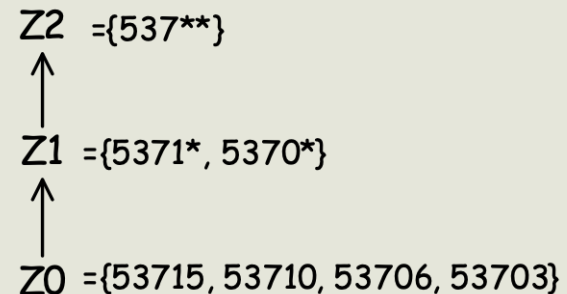
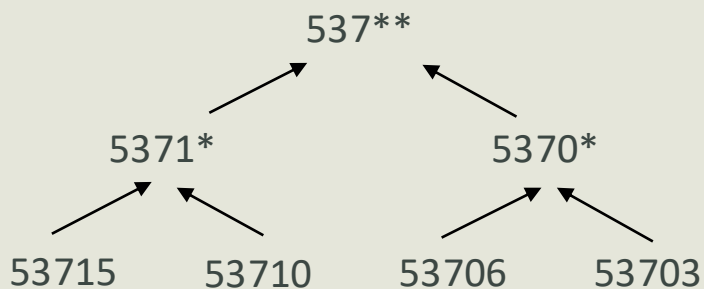
A **domain generalization hierarchy** DGH_D of an attribute A is a partial order on the set of domains $Dom_A = \{D_0, \dots, D_n\}$ satisfying the following conditions:

1. Each domain D_i has at most one direct generalized domain
2. All maximal elements of Dom are singletons (to ensure that all values in each domain can be possibly generalized to a single value)



Value generalization hierarchies

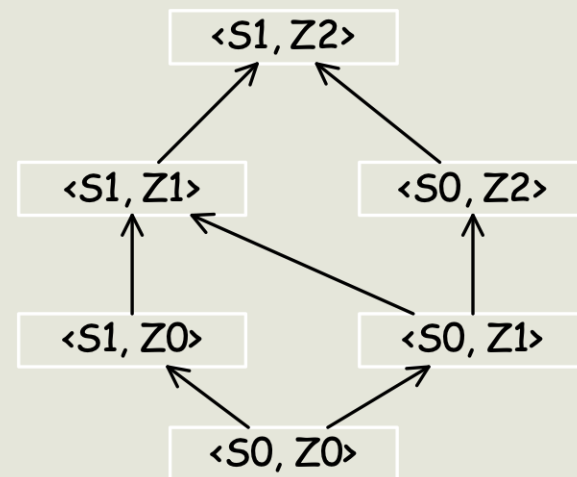
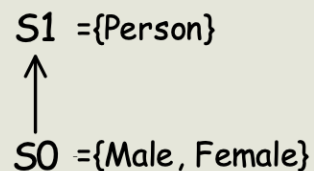
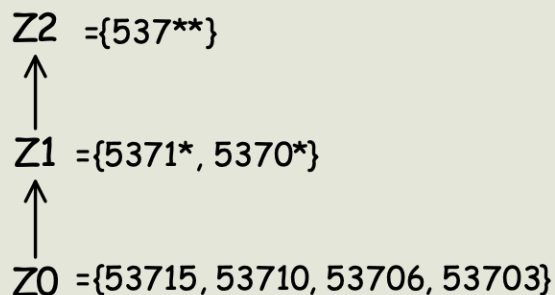
- A **value generalization relationship** associates with each value v_i of the domain D_i a unique value v_j in domain D_j , where D_j is a direct generalization of D_i
- This relationship implies the existence, for each domain D , of a **value generalization hierarchy**, denoted VGH_D
- VGH_D is a **tree**, where the leaves are the values in D and the root (i.e., the most general value) is the value in the maximum element in DGH_D



Generalization lattice

- Given a domain tuple $DT = \langle D_{A1}, \dots, D_{An} \rangle$ such that D_{Ai} in Dom_{Ai} , the **domain generalization hierarchy** of DT is

$$\text{DGH}_{DT} = \text{DGH}_{D_{A1}} \times \dots \times \text{DGH}_{D_{An}}$$
- DGH_{DT} defines a **lattice** whose minimal element is DT



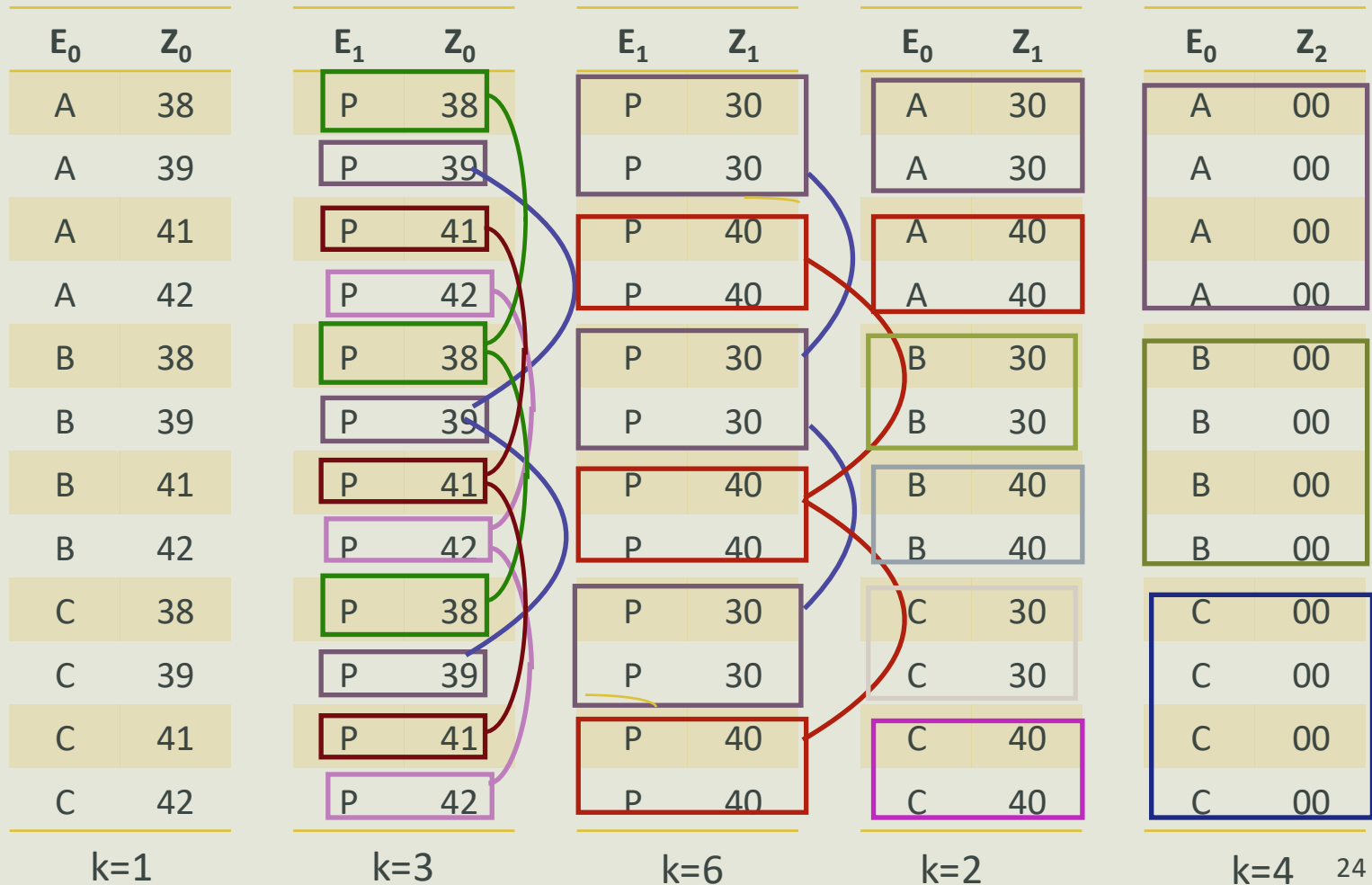
Generalized table with suppression

Given a set of attributes $A = \{A_1, \dots, A_n\}$ and two tables T_i and T_j defined on A , table T_j is a **generalization (with tuple suppression)** of table T_i , denoted $T_i \leq T_j$, iff:

1. The domain of each attribute A_x in T_j is equal to or a generalization of the domain of A_x in T_i
2. Each tuple t_j in T_j has a corresponding tuple t_i in T_i such that for each attribute A_x , $t_j[A_x]$ is equal to or a generalization of $t_i[A_x]$

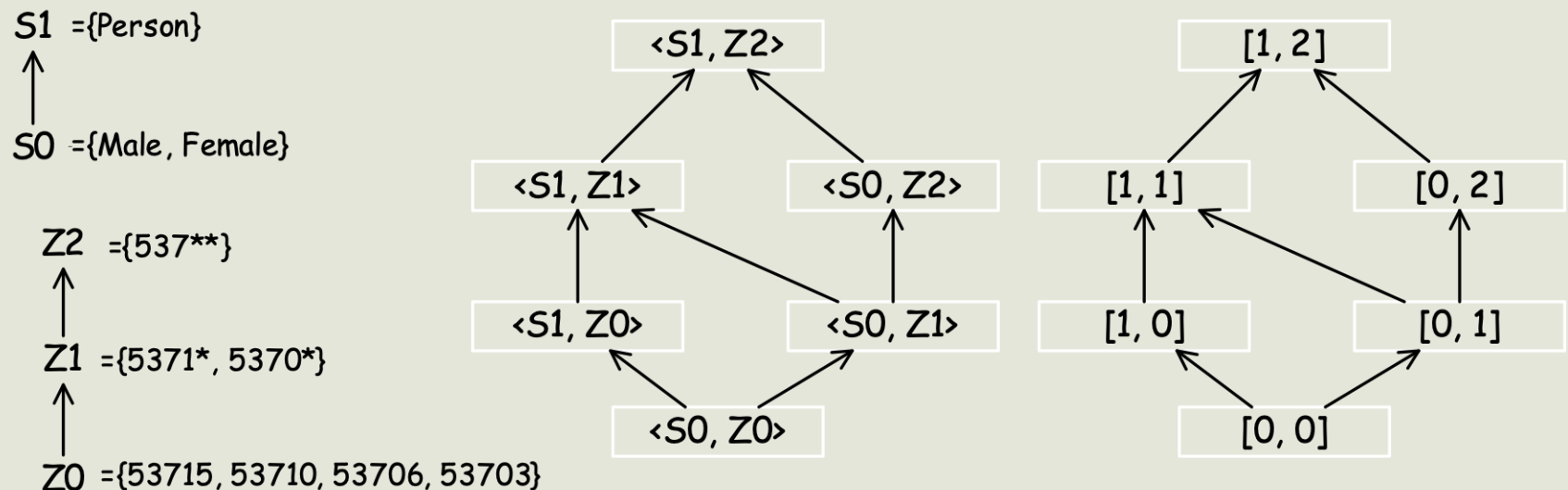
Not all generalizations are the same!

Examples

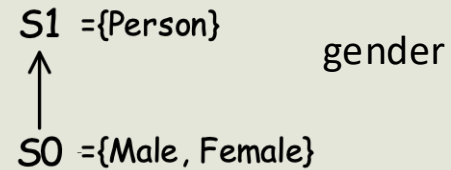
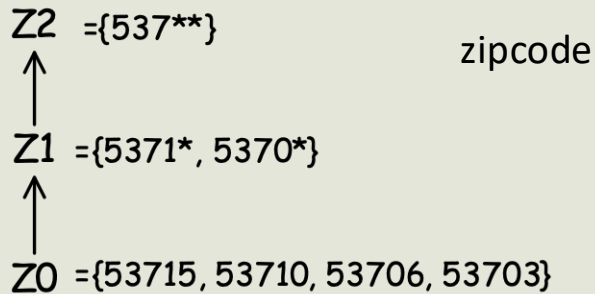


Distance vector

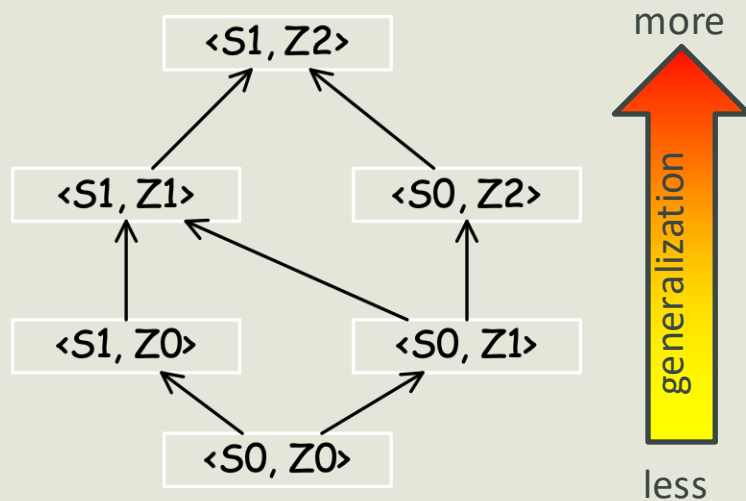
Let T_i and T_j be two tables defined on the same set of attributes $A = \{A_1, \dots, A_n\}$ such that $T_i \leq T_j$. The **distance vector** of T_j from T_i is the vector $DV_{ij} = [d_1, \dots, d_n]$ where each d_x is the length of the unique path between $\text{dom}(A_x, T_i)$ and $\text{dom}(A_x, T_j)$ in the domain generalization hierarchy DGH_{Dx} .



Generalization criteria (intuition)

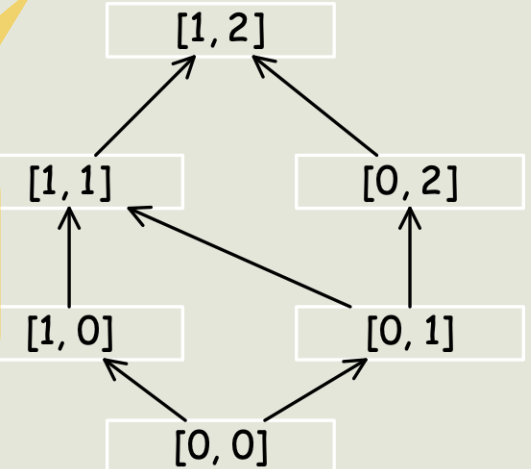


construct the **generalization lattice** for the entire QI set



objective
find the minimum generalization that satisfies k-anonymity

i.e., maximize **utility** by finding minimum distance vector with k-anonymity



k-minimal generalization with suppression

Let T_i and T_j be two tables such that $T_i \leq T_j$ and let MaxSup be the specified threshold of acceptable suppression.

T_j is a **k-minimal generalization** of T_i iff

1. T_j satisfies k -anonymity

2. Suppression is minimal

$\forall T_x: T_i \leq T_x, DV_{ix} = DV_{ij}, T_x \text{ is } k\text{-anonymous} \Rightarrow |T_j| \geq |T_x|$

T_j has the same properties as T_x tables but has a higher cardinality

3. $|T_j| - |T_i| \leq \text{MaxSup}$

4. $\forall T_x: T_i \leq T_x, T_x \text{ satisfies 1, 2 and 3} \Rightarrow DV_{ix} \geq DV_{ij}$

*Among tables T_x
 T_j has a lower number of generalizations*

2-minimal generalization (examples)

Ethnicity: E_0	ZIP: Z_0
Asian	94142
Asian	94141
Asian	94139
Asian	94139
Asian	94139
AfrAm	94138
AfrAm	94139
Caucasian	94139
Caucasian	94141

Ethnicity: E_1	ZIP: Z_0
Person	94141
Person	94139
Person	94139
Person	94139
Person	94139
Person	94139
Person	94141

DV=[1,0]

Ethnicity: E_0	ZIP: Z_1
Asian	9414*
Asian	9414*
Asian	9413*
Asian	9413*
Asian	9413*
AfrAm	9413*
AfrAm	9413*

DV=[0,1]

Classification of k -anonymity techniques

Generalization and suppression can be applied at different levels of granularity

- Generalization can be applied
 - **at single column level**: a generalization step generalizes all the values in the column
 - **at single cell level**: for a specific column, the table may contain values at different generalization levels
- Suppression can be applied
 - **at row level**: suppression operation removes a whole tuple
 - **at column level**: suppression operation obscures all the values of a column
 - **at single cell level**: only certain cells of a given tuple/attribute can be wiped out

Algorithms for k -anonymity

- The problem of finding minimal k -anonymous tables, with attribute generalization and tuple suppression, is **NP-hard**
- The majority of exact algorithms proposed in literature have computational time exponential in the number of the attributes composing the quasi-identifier
- When the number $|QI|$ of attributes in the quasi-identifier is small compared with the number n of tuples in the private table, these exact algorithms with attribute generalization and tuple suppression are practical

Samarati's algorithm

P. Samarati: Protecting Respondents' Identities in Microdata Release. IEEE Trans. Knowl. Data Eng. 13(6): 1010-1027 (2001)

- Each path in DGH_{DT} represents a generalization strategy
- The algorithm is based on **locally minimal generalization** (the lowest node of each path satisfying k -anonymity)
- Other properties
 1. each k -minimal generalization is locally minimal with respect to a path (the opposite is not true)
 2. going up in the hierarchy, the number of tuples that must be removed to guarantee k -anonymity decreases
- If there is no solution that guarantees k -anonymity suppressing less than $MaxSup$ tuples at height h , there cannot exist a solution with height lower than h that guarantees it

Samarati's algorithm

P. Samarati: Protecting Respondents' Identities in Microdata Release. IEEE Trans. Knowl. Data Eng. 13(6): 1010-1027 (2001)

- The algorithm adopts a binary search on the lattice of distance vectors
 1. evaluate solutions at height $\lfloor h/2 \rfloor$
 2. if there exists at least a solution satisfying k -anonymity
 - then evaluate solutions at height $\lfloor h/4 \rfloor$
 - otherwise evaluate solutions at height $\lfloor 3h/4 \rfloor$
 3. until the algorithm reaches the lowest height for which there is a distance vector that satisfies k -anonymity
- To reduce the computational cost, it adopts a **distance vector matrix** that avoids the explicit computation of each generalized table

Example $h=1$ ($k=2$ and $\text{MaxSup}=2$)

- Compute first solution at height 1 ($GT_{[1,0]}$ and $GT_{[0,1]}$)

	Ethn.: E_1	Zip: Z_0
suppressed	person	94142(*)
	person	94141
	person	94139
	person	94139
	person	94139
suppressed	person	94138(*)
	person	94139
	person	94139
	person	94141

	Ethn.: E_1	Zip: Z_0
	asian	9414*
	asian	9414*
	asian	9413*
	asian	9413*
	asian	9413*
	AfrAm	9413*
	AfrAm	9413*
suppressed	caucas(*)	9413*
suppressed	caucas(*)	9414*

- Both the generalized tables satisfy 2-anonymity

Example $h=0$ ($k=2$ and $\text{MaxSup}=2$)

- Compute solution at height 0 ($\text{GT}_{[0,0]}$)

Ethn.: E_0	Zip: Z_0
asian	94142
asian	94141
asian	94139
asian	94139
asian	94139
AfrAm	94138
AfrAm	94139
Caucas.	94139
Caucas.	94141

- The above table does not satisfy 2-anonymity

Incognito algorithm

K. LeFevre, D.J. DeWitt, R. Ramakrishnan:
Incognito: Efficient Full-Domain K-Anonymity.
SIGMOD Conference 2005: 49-60.

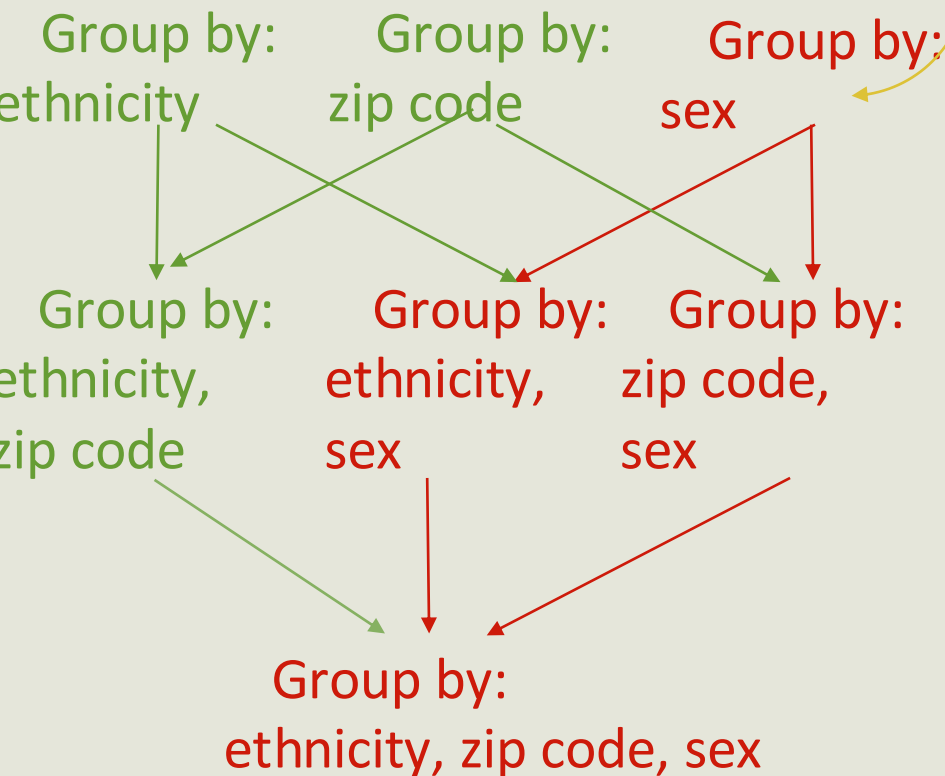
k -anonymity with respect to a proper subset of QI is a necessary (not sufficient) condition for k -anonymity with respect to QI

- **Iteration 1:** check k -anonymity for each attribute in QI, discarding generalizations that do not satisfy k -anonymity
- **Iteration 2:** combine the remaining generalizations in pairs and check k -anonymity for each couple obtained
- ...
- **Iteration i :** combine all the i -uples of attributes, obtained combining generalizations that satisfied k -anonymity at iteration $i - 1$. Discard non k -anonymous solutions
- Iteration $|QI|$ returns the final result

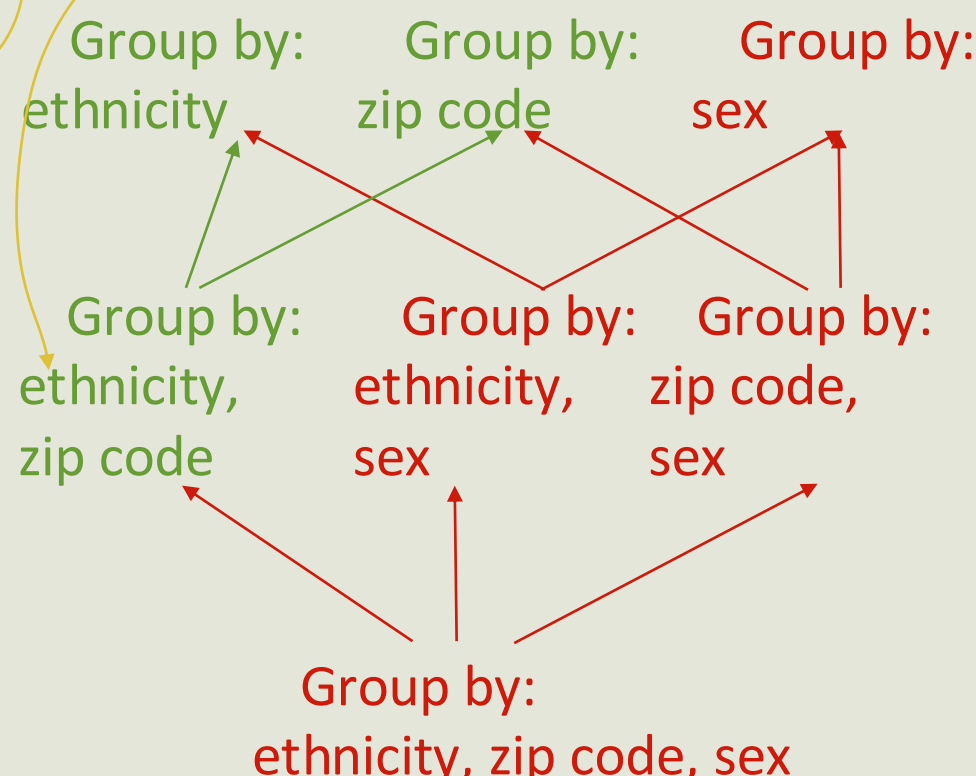
Incognito adopts a bottom-up approach for the visit of DGHs

k -anonymity with respect to a proper subset of QI is a necessary (not sufficient) condition for k -anonymity with respect to QI

Monotonicity from top to bottom:
if grouped table **does not** have k -anonymity,
neither with **subgroups** (QI with more attributes)



Monotonicity from bottom to up:
if a grouped table **has** k -anonymity,
also with **supergroups** (QI with less attributes)



k -anonymity with respect to a proper subset of QI is a necessary (not sufficient) condition for k -anonymity with respect to QI

- Assume $k=2$

Group by: ethnicity, zip code

Ethnicity: E_0	ZIP: Z_0
Asian	94142
Asian	94142
Asian	94139
Asian	94139
Asian	94139
AfrAm	94138
AfrAm	94138
Caucasian	94142
Caucasian	94142

2-anonymous:

Group by: ethnicity

Ethnicity: E_0	ZIP: Z_0
Asian	94142
Asian	94142
Asian	94139
Asian	94139
Asian	94139
AfrAm	94138
AfrAm	94138
Caucasian	94142
Caucasian	94142

2-anonymous:

Group by: zip code

Ethnicity: E_0	ZIP: Z_0
Asian	94142
Asian	94142
Asian	94139
Asian	94139
Asian	94139
AfrAm	94138
AfrAm	94138
Caucasian	94142
Caucasian	94142

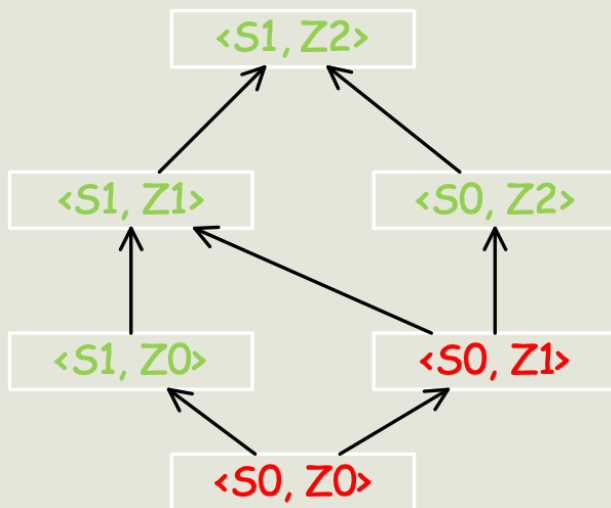
2-anonymous:

&

Incognito algorithm

Exploits **monotonicity properties** regarding frequency of tuples in lattice

- reminiscent of OLAP hierarchies and frequent itemset mining



(I) **generalization property** ($\sim$ rollup)

if at some node k -anonymity holds, then it also holds for any ancestor node

e.g., $\langle S1, Z0 \rangle$ is k -anonymous and, thus, so is $\langle S1, Z1 \rangle$ and $\langle S1, Z2 \rangle$

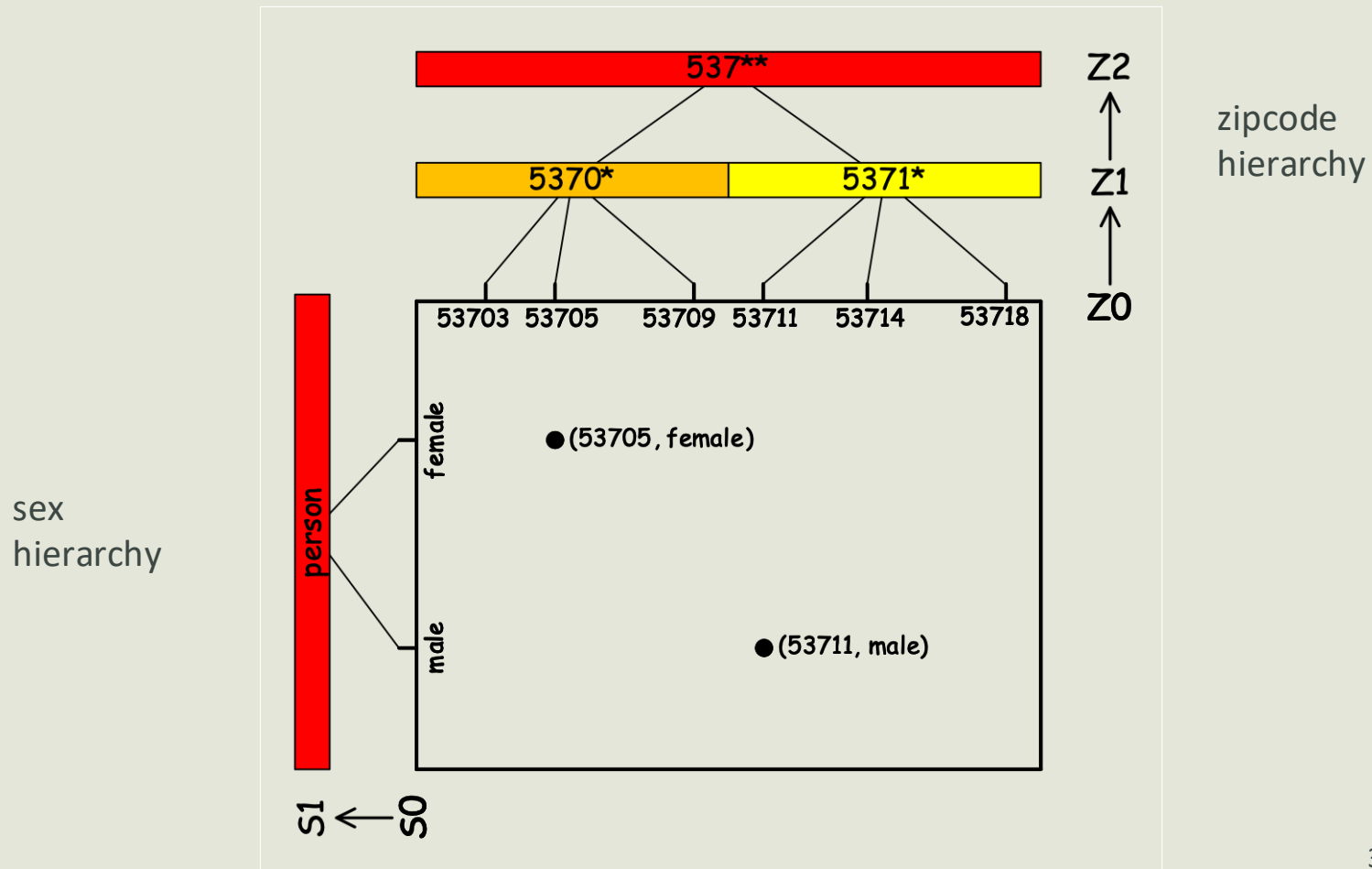
(II) **subset property** ($\sim$ apriori)

if for a set of QI attributes k -anonymity doesn't hold then it doesn't hold for any of its supersets

e.g., $\langle S0, Z0 \rangle$ is not k -anonymous and, thus $\langle S0, Z0, B0 \rangle$ and $\langle S0, Z0, B1 \rangle$ cannot be k -anonymous

Incognito considers sets of QI attributes of increasing cardinality ($\sim$ apriori) and prunes nodes in the lattice using the two properties above

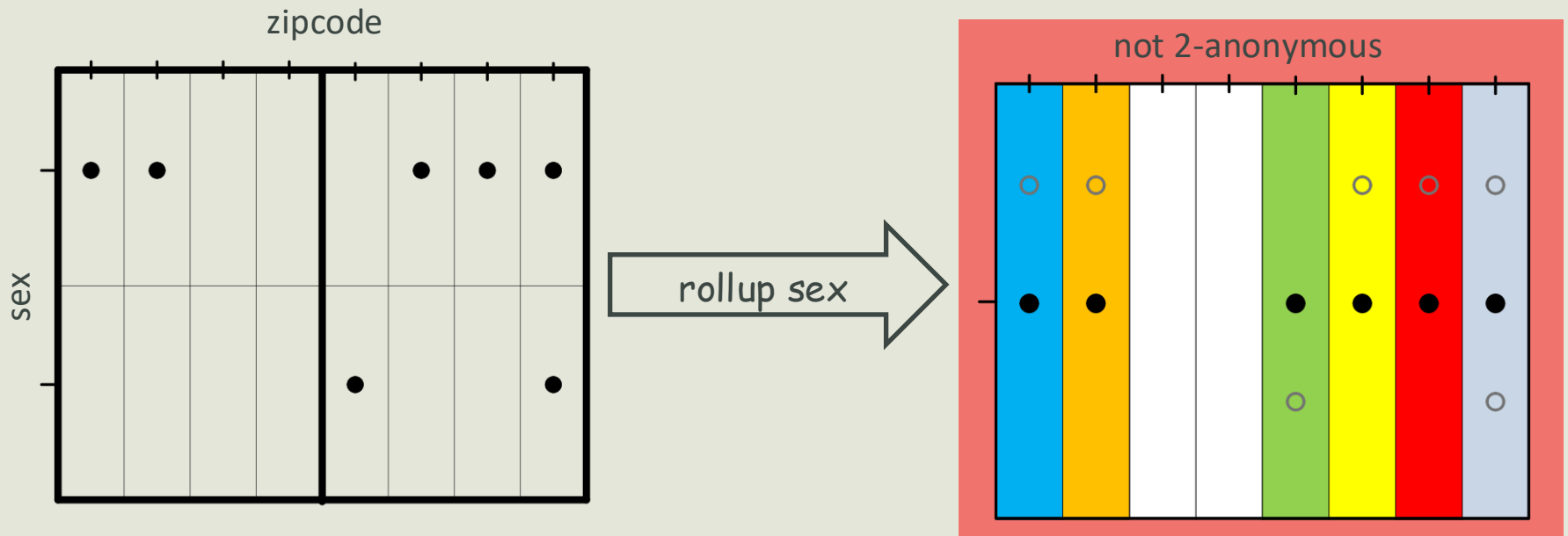
Incognito in the domain space



Example ($k=2$)

Incognito example

2 QI attributes, 7 tuples, hierarchies shown with bold lines



Example ($k=2$)

incognito example

2 QI attributes, 7 tuples, hierarchies shown with bold lines

	zipcode							
sex	•	•				•	•	•
						•		•

rollup zipcode

